Fastsae: High-Performance Frequentist and Bayesian Small Area Estimation in R

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Abstract Small area estimation (SAE) plays a pivotal operational role in official statistics, enabling the production of reliable disaggregated socioeconomic, health, and demographic indicators when domain-specific sample sizes are small or zero. However, practitioners in national statistical offices (NSOs) face a triple computational and operational barrier: frequentist spatial and spatio-temporal models with bootstrap mean squared error (MSE) estimation suffer from severe memory bottlenecks and slow runtimes in interpreted languages; non-Gaussian indicators (such as rates, skewed expenditures, and counts) traditionally require computationally demanding Markov chain Monte Carlo (MCMC) algorithms that impede routine production; and model estimates often fail to aggregate to published higher-level totals, violating institutional consistency. We introduce fastsae, an R package engineered for high-performance frequentist and Bayesian small area estimation. On the frequentist side, fastsae implements optimized C++17 routines via RcppArmadillo paired with OpenMP multi-threading, accelerating the classical Fay-Herriot, spatial Fay-Herriot, spatio-temporal Fay-Herriot, and unit-level Battese-Harter-Fuller models alongside their analytical and bootstrap MSE estimators. On the Bayesian side, fastsae provides a unified empirical best prediction (EBP) interface across six distribution families (Gaussian, Beta, Gamma, Binomial, Poisson, Negative Binomial), fitted by Integrated Nested Laplace Approximations (INLA) by default or by MCMC in Python (JAX/NumPyro) through interchangeable engine objects. Crucially, fastsae introduces exact survey dispersion mappings that respect design-based direct sampling variances and accommodates modern spatial priors (BYM, scaled BYM2, Leroux CAR, and SLM) as well as Knorr-Held space-time interaction dynamics. To bridge estimation and dissemination, the package features a two-stage hierarchical benchmarking engine for simultaneous provincial and national calibration, an automated reliability classification engine aligned with NSO standards, multi-model concordances, automated ggplot2 spatial mapping, and multi-sheet spreadsheet exports. Extensive benchmarks and finite population Monte Carlo simulations demonstrate large computational speedups at 1,000 domains (1,263x for Fay-Herriot and 67x for spatial Fay-Herriot relative to sae, 217x for spatial Beta relative to Stan-based MCMC) and reductions of more than 99% in allocated memory, without loss of inferential precision.

1 Introduction

In modern evidence-based governance, policy interventions are increasingly decentralized. Monitoring the United Nations Sustainable Development Goals (SDGs), allocating municipal fiscal transfers, targeting social safety net subsidies, and responding to localized disease outbreaks all require dependable statistical indicators at highly disaggregated geographic administrative levels (Rao and Molina, 2015; Pfeffermann, 2013). However, national statistical offices (NSOs) face structural sample size limitations. Large-scale household surveys—such as the American Community Survey (ACS) in the United States, the European Union Statistics on Income and Living Conditions (EU-SILC), and the National Socioeconomic Survey (Susenas) in Indonesia—are powered to yield acceptable sampling errors only at national or broad regional levels. When partitioned into sub-domains (e.g., municipalities, districts, or health zones), domain-specific sample sizes become sparse or zero, causing direct survey estimators to suffer from high sampling variance and unacceptable coefficients of variation (Burgard et al., 2020; Pratesi, 2016).

Small area estimation (SAE) overcomes these limitations by "borrowing strength'' across domains and time periods via linking models that combine survey responses with auxiliary administrative registers, census aggregates, or geospatial layers (Rao and Molina, 2015). The cornerstone of this literature remains the area-level Fay-Herriot (FH) model (Fay and Herriot, 1979), extended to account for spatial dependencies via Simultaneous Autoregressive (SAR) structures (Petrucci and Salvati, 2006; Pratesi and Salvati, 2008) and temporal dynamics via panel autoregressive processes (Marhuenda et al., 2013). In the presence of micro-level auxiliary data, the nested error regression model of Battese et al. (1988) (BHF) provides unit-level empirical best linear unbiased predictors (EBLUP).

Despite these theoretical advances, operationalizing small area models within production environments in NSOs has been hindered by three structural bottlenecks: %

1. **Frequentist Computational and Memory Bottlenecks**: Existing implementations of spatial and spatio-temporal Fay-Herriot models rely on interpreted languages. Estimating model variance components via Restricted Maximum Likelihood (REML) requires repeated inversions of $D \times D$ or $DT \times DT$ covariance matrices. When evaluating Mean Squared Error (MSE)

Feature / Package	fastsae	sae	emdi	tipsae	mcmcsae	SUMMER
Computational Engine	C++17 / INLA	Pure R	R / nlme	Stan HMC	C++ / MCMC	INLA
Frequentist EBLUP (FH/SFH)	✓	✓	✓	×	×	×
Spatio-Temporal Panel EBLUP	✓	✓	×	×	×	×
Unit-Level BHF Model	✓	✓	✓	×	×	×
OpenMP Parallel Bootstrap	✓	×	×	×	×	×
Non-Gaussian Likelihoods	6 Families	×	Box-Cox	Beta only	4 Families	3 Families
Deterministic Bayesian EBP	✓	×	×	×	×	✓
Exact Survey Dispersion Match	✓	×	×	✓	×	×
Modern Spatial Priors (BYM2/Leroux)	✓	×	×	Besag	Besag	BYM2
Two-Stage Hierarchical Calibration	✓	×	×	×	×	×
Official Statistics Reliability Flags	✓	×	×	×	×	×
Automated Production Mapping	✓	×	✓	×	×	✓
Multi-Sheet Dissemination Export	✓	×	✓	×	×	×

Table 1: Comparative landscape of Small Area Estimation packages in the R ecosystem.

through parametric or non-parametric bootstrap resampling (González-Manteiga et al., 2008), runtimes escalate to hours. Crucially, memory allocation for dense covariance arrays grows quadratically ($O(D^2)$ to $O(D^2T^2)$), triggering out-of-memory (OOM) crashes on large national grids ($D \geq 500$).

2. **Distributional Inflexibility and MCMC Friction in Bayesian SAE**: Real-world welfare indicators are rarely Gaussian. Rates and proportions (poverty headcounts, unemployment, stunting) are strictly bounded in $(0, 1)$, while expenditures exhibit severe positive skewness (Gamma), and demographic events represent discrete counts (Poisson, Negative Binomial) (Janicki, 2020; Fabrizi et al., 2011; Boubeta et al., 2016). Under non-Gaussian likelihoods, linear EBLUP is invalid, requiring Empirical Best Prediction (EBP) within a Hierarchical Bayesian (HB) framework. Currently, Bayesian SAE software relies on Markov Chain Monte Carlo (MCMC) sampling via **Stan** or **JAGS** (De Nicolò and Gardini, 2022). MCMC sampling requires lengthy runtimes, complex convergence diagnostics ($\hat{R}$, trace plots), and often suffers from divergent transitions on complex spatial graphs (Simpson et al., 2017).
3. **The Institutional Benchmarking and Dissemination Gap**: In official statistics, model-based small area estimates must numerically harmonize with published higher-level totals (e.g., provincial and national direct estimates) to preserve institutional credibility and prevent contradictory policy decisions (You and Rao, 2002; Wang et al., 2008; Stefan and Hidiroglou, 2020). Existing packages rarely offer two-stage hierarchical benchmarking. Furthermore, software ecosystems typically end at model fitting, omitting essential dissemination steps such as automated reliability tagging based on official Relative Standard Error (RSE) thresholds, model concordance evaluation, and publication-ready spatial choropleth mapping.

As summarized in Table 1, the current software landscape is compartmentalized. Foundational toolkits such as **sae** (Molina and Marhuenda, 2015) are implemented in interpreted R, lack multithreading, and are restricted to Gaussian linear frameworks. Packages such as **emdi** (Kreutzmann et al., 2019) offer visualizations but lack spatio-temporal dynamics and non-Gaussian spatial Bayesian likelihoods. Specialized packages such as **tipsae** (De Nicolò and Gardini, 2022) implement Beta regression but depend on MCMC runtimes that scale poorly for panel graphs. Toolkits such as **mcmcsae** (Boonstra, 2022) remain simulation-based, while **SUMMER** (Li et al., 2020) is tailored to demographic survey mortality rather than general survey design-variance matching.

To resolve these limitations, this paper introduces **fastsae**. Specifically, this paper makes four concrete, testable contributions: %

1. **Compiled C++17 Core with OpenMP Acceleration**: We engineer an optimized linear algebra core in C++17 via **RcppArmadillo** that speeds up classical, spatial, and spatio-temporal Fay-Herriot estimation with analytical MSE by $1,263\times$, $67\times$, and $24\times$ relative to **sae** (and the classical model by $42,874\times$ relative to **emdi**) at $D = 1,000$, while reducing allocated memory from gigabytes to a few megabytes. Sparse Cholesky factorizations and structured Woodbury identities avoid every dense $D \times D$ (or $DT \times DT$) inverse.
2. **Unified Deterministic Bayesian Hierarchical Modeling via INLA with Exact Survey Matching**: We formulate a Hierarchical Bayes interface (`hb_area()`) spanning six probability distributions (Gaussian, Beta, Gamma, Binomial, Poisson, Negative Binomial) powered by Integrated Nested Laplace Approximations (Rue et al., 2009), with an optional MCMC engine in Python (JAX/NumPyro) selected through the same interface (`engine = inla_engine()` or `engine = jax_engine()`). It embeds exact survey variance mappings—notably the Janicki (2020) Beta dispersion parameterization and Gamma CV matching—achieving $117\times$ to $530\times$ speedups over

Stan-based MCMC at the largest benchmarked sizes while maintaining near-perfect inferential concordance ($r = 0.9993$).

3. **Two-Stage Hierarchical Benchmarking Engine**: We implement a modular calibration engine (`benchmark_sae()`) that simultaneously enforces consistency across multi-level administrative hierarchies (Level 0 National, Level 1 Provincial, Level 2 District) via ratio, difference, variance-weighted optimal, and logit-scale adjustments.
4. **End-to-End Official Statistics Dissemination Suite**: We provide turnkey post-estimation workflows, including automated reliability tagging (`add_reliability_flags()`) based on official NSO thresholds, model concordance diagnostics (`compare_sae()`), publication-ready choropleth mapping (`map_sae()`), and multi-sheet spreadsheet exports (`export_sae()`).

2 Methodological framework

2.1 Frequentist small area models

Classical Fay-Herriot model

Let m denote the number of small areas indexed by $i = 1, \ldots, m$. The classical area-level model (Fay and Herriot, 1979) is defined by: %

$$y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_i + e_i, \quad u_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_u^2), \quad e_i \overset{\text{ind}}{\sim} \mathcal{N}(0, D_i), \tag{1}$$

% where y_i is the design-based direct estimator of the target indicator θ_i, $\boldsymbol{x}_i$ is a $p \times 1$ vector of auxiliary covariates, D_i is the known design variance, and u_i is the area random effect. The Empirical Best Linear Unbiased Predictor (EBLUP) is given by: %

$$\hat{\theta}_i = \hat{\gamma}_i y_i + (1 - \hat{\gamma}_i) \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}, \quad \hat{\gamma}_i = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + D_i}, \tag{2}$$

% where $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \hat{\boldsymbol{V}}^{-1} \boldsymbol{X})^{-1} \boldsymbol{X}^\top \hat{\boldsymbol{V}}^{-1} \boldsymbol{y}$ with $\boldsymbol{V} = \text{diag}(\sigma_u^2 + D_1, \ldots, \sigma_u^2 + D_m)$. Mean squared error is evaluated via the second-order Prasad-Rao approximation (Prasad and Rao, 1990): %

$$\text{MSE}(\hat{\theta}_i) \approx g_{1i}(\hat{\sigma}_u^2) + g_{2i}(\hat{\sigma}_u^2) + 2g_{3i}(\hat{\sigma}_u^2). \tag{3}$$

Spatial Fay-Herriot model

To accommodate spatial dependence across contiguous domains, Petrucci and Salvati (2006) and Pratesi and Salvati (2008) modeled the area random effects $\boldsymbol{u} = (u_1, \ldots, u_m)^\top$ via a Simultaneous Autoregressive (SAR) process: %

$$\boldsymbol{u} = \rho \boldsymbol{W} \boldsymbol{u} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}_m(\boldsymbol{0}, \sigma_1^2 \boldsymbol{I}_m), \tag{4}$$

% where ρ is the spatial autoregressive parameter and $\boldsymbol{W}$ is a row-standardized spatial proximity matrix. The marginal covariance matrix is $\boldsymbol{V}(\sigma_1^2, \rho) = \boldsymbol{G}(\sigma_1^2, \rho) + \boldsymbol{D}$, where $\boldsymbol{G}(\sigma_1^2, \rho) = \sigma_1^2 [(\boldsymbol{I}_m - \rho \boldsymbol{W})^\top (\boldsymbol{I}_m - \rho \boldsymbol{W})]^{-1}$. The spatial EBLUP is: %

$$\hat{\boldsymbol{\theta}} = \boldsymbol{X} \hat{\boldsymbol{\beta}} + \hat{\boldsymbol{G}} \hat{\boldsymbol{V}}^{-1} (\boldsymbol{y} - \boldsymbol{X} \hat{\boldsymbol{\beta}}). \tag{5}$$

% The SAR process is defined on all domains of $\boldsymbol{W}$, including unsampled ones, which are predicted by spatial kriging. Variance components (σ_1^2, ρ) are estimated by REML or ML Fisher scoring with $|\rho| \leq 0.999$, falling back to a constrained (projected) Fisher scoring when the classical iteration ends at a negative variance. All quantities are obtained from sparse Cholesky factorizations of $\boldsymbol{A} = (\boldsymbol{I} - \rho \boldsymbol{W})^\top (\boldsymbol{I} - \rho \boldsymbol{W})$ and $\boldsymbol{A} + \sigma_1^2 \boldsymbol{E}^\top \boldsymbol{D}^{-1} \boldsymbol{E}$ through the Woodbury identity, so no dense $m \times m$ inverse is formed. Mean squared error uses the same second-order approximation $g_1 + g_2 + 2g_3 - g_4$ as **sae** (Molina and Marhuenda, 2015), or parametric and non-parametric bootstrap.

Spatio-temporal Fay-Herriot model

When small area data are collected across longitudinal waves $t = 1, \ldots, T$, borrowing strength across space and time yields substantial efficiency gains (Marhuenda et al., 2013): %

$$y_{it} = \boldsymbol{x}_{it}^\top \boldsymbol{\beta} + u_{1,it} + u_{2,it} + e_{it}, \tag{6}$$

% where $\boldsymbol{u}_1 = \rho_1 \boldsymbol{W}\boldsymbol{u}_1 + \boldsymbol{\epsilon}_1$ is a spatial SAR effect that is constant over time within each area ($u_{1,it} = u_{1,i}$) and $u_{2,it} = \rho_2 u_{2,i,t-1} + \epsilon_{2,it}$ follows an AR(1) temporal process with $|\rho_2| < 1$, as in Marhuenda et al. (2013). In **fastsae**, the covariance $\boldsymbol{V} = \boldsymbol{A} + \boldsymbol{Z}\boldsymbol{V}_{u_1}\boldsymbol{Z}^\top$ (with $\boldsymbol{A}$ block diagonal in $T \times T$ blocks) is never formed: REML scores, Fisher information, and the analytical MSE ($g_1 + g_2 + 2g_3 - g_4$) are computed from $T \times T$, $D \times D$, and low-rank $DT \times (D + p)$ pieces via the Woodbury identity.

Unit-level Battese-Harter-Fuller model

When unit microdata are accessible alongside population auxiliary totals, **fastsae** implements the nested error regression model (Battese et al., 1988): %

$$y_{dj} = \boldsymbol{x}_{dj}^\top \boldsymbol{\beta} + u_d + e_{dj}, \quad d = 1, \ldots, D, \quad j = 1, \ldots, n_d, \tag{7}$$

% yielding EBLUP predictors $\hat{\bar{Y}}_d = f_d \bar{y}_d + (1 - f_d)(\bar{\boldsymbol{X}}_{d,r}^\top \hat{\boldsymbol{\beta}} + \hat{\gamma}_d(\bar{y}_d - \bar{\boldsymbol{x}}_d^\top \hat{\boldsymbol{\beta}}))$, where $f_d = n_d / N_d$.

2.2 Bayesian non-Gaussian hierarchical EBP via INLA

Latent Gaussian models

For non-Gaussian indicators, **fastsae** formulates small area estimation within the Latent Gaussian Model (LGM) framework (Rue et al., 2009): %

$$y_i \mid \eta_i, \boldsymbol{\theta} \sim \pi(y_i \mid \eta_i, \boldsymbol{\theta}), \quad \eta_i = g(\mu_i) = \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_i + v_i, \tag{8}$$

% where $g(\cdot)$ is a monotone link function, $v_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \tau_v^{-1})$ is an unstructured effect, and $\boldsymbol{u} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}^{-1}(\boldsymbol{\theta}_s))$ is a spatial Gaussian Markov Random Field (GMRF). By default, **fastsae** utilizes the Full Laplace approximation (`strategy = "laplace"`) rather than the simplified Laplace expansion, guaranteeing high analytical fidelity even near boundary domains.

Exact survey dispersion and variance matching

To preserve design-based sampling properties in a Bayesian GLMM, **fastsae** injects distribution-specific dispersion parameters: %

1. **Gaussian**: Precision is fixed to the inverse design variance: $\tau_i = 1/D_i$.
2. **Beta**: For proportions $y_i \in (0, 1)$ with logit link $\text{logit}(\mu_i) = \eta_i$, the Beta variance is $\text{Var}(y_i) = \mu_i(1 - \mu_i)/(1 + \phi_i)$. Equating this to design variance D_i yields the Janicki (2020) dispersion:
$$\phi_i = \max\left(\frac{y_i(1 - y_i)}{D_i} - 1,\, 1\right). \tag{9}$$
3. **Gamma**: For positive skewed expenditures $y_i > 0$ with log link $\log(\mu_i) = \eta_i$, equating $\text{Var}(y_i) = \mu_i^2 / s_i$ to D_i yields the shape precision parameter:
$$s_i = \frac{y_i^2}{D_i} = \frac{1}{\text{CV}_i^2}. \tag{10}$$
4. **Binomial, Poisson, and Negative Binomial**: Domain sample sizes n_i map to `Ntrials`, and population exposures E_i map to the log-link offset $\log(E_i)$.

Spatial priors and space-time interactions

The spatial random field $\boldsymbol{u}$ supports five graph topologies: Besag ICAR (Besag et al., 1991), scaled BYM2 with Penalized Complexity (PC) priors (Riebler et al., 2016; Simpson et al., 2017), Leroux CAR with exact precision $\boldsymbol{Q} = \tau[(1 - \rho)\boldsymbol{I} + \rho(\text{diag}(\boldsymbol{d}) - \boldsymbol{W})]$ (Leroux et al., 2000), and the Spatial Lag Model (SLM). For panel observations across $T \geq 2$, temporal trends support random walks (RW1, RW2) and AR(1) dynamics. Space-time interactions can be specified as additive, domain-specific, separable, or according to the four Knorr-Held (2000) interaction types ($\boldsymbol{Q}_{st} = \boldsymbol{Q}_s \otimes \boldsymbol{Q}_t$).

2.3 Two-stage hierarchical benchmarking calibration

In official statistics, model-based small area estimates $\hat{\theta}_{di}$ for domain i in province/region g must aggregate to officially published higher-level targets (Rao and Molina, 2015; Stefan and Hidiroglou, 2020). Let w_{di} denote domain population weights such that $W_g = \sum_{i \in g} w_{di}$ is the provincial total weight, and $W_{\text{nat}} = \sum_g W_g$ is the national total weight. Let T_g denote the provincial target and T_{nat} denote the national aggregate target.

fastsae implements a two-stage hierarchical benchmarking calibration: %

1. **Stage 1: Harmonization of Provincial Targets to National Benchmark**: When initial provincial targets T_g (or initial aggregated model estimates $\hat{\tilde{\theta}}_g = \sum_{i \in g}(w_{di}/W_g)\hat{\theta}_{di}$) do not sum to the national target T_{nat}, calibrated provincial targets $\tilde{T}_g$ are computed to satisfy $\sum_g (W_g/W_{\text{nat}})\tilde{T}_g = T_{\text{nat}}$.
2. **Stage 2: Domain-Level Calibration within Provinces**: Domain estimates within province g are calibrated to satisfy $\sum_{i \in g}(w_{di}/W_g)\hat{\theta}^{\text{BM}}_{di} = \tilde{T}_g$.

% Across both stages, **fastsae** supports four calibration mechanisms: %

- **Ratio Benchmarking** (You and Rao, 2002): $\hat{\theta}^{\text{BM}}_{di} = \hat{\theta}_{di}\left(\tilde{T}_g / \sum_{j \in g} \tilde{w}_{dj}\hat{\theta}_{dj}\right)$.
- **Difference Benchmarking** (Rao and Molina, 2015): $\hat{\theta}^{\text{BM}}_{di} = \hat{\theta}_{di} + \left(\tilde{T}_g - \sum_{j \in g} \tilde{w}_{dj}\hat{\theta}_{dj}\right)$.
- **Variance-Weighted Optimal Benchmarking** (Datta et al., 2011; Steorts and Ghosh, 2014): Adjustments minimize $\sum_i (\hat{\theta}^{\text{BM}}_{di} - \hat{\theta}_{di})^2 / \text{MSE}(\hat{\theta}_{di})$ subject to the benchmark constraint, which allocates them in proportion to $\tilde{w}_{di}\,\text{MSE}(\hat{\theta}_{di})$:

$$\hat{\theta}^{\text{BM}}_{di} = \hat{\theta}_{di} + \tilde{w}_{di}\text{MSE}(\hat{\theta}_{di}) \left(\sum_{j \in g} \tilde{w}^2_{dj}\text{MSE}(\hat{\theta}_{dj})\right)^{-1} \left(\tilde{T}_g - \sum_{j \in g} \tilde{w}_{dj}\hat{\theta}_{dj}\right). \tag{11}$$

- **Logit-Scale Benchmarking**: For bounded proportions in $(0,1)$, linear adjustments can push estimates outside $[0,1]$. **fastsae** solves for an additive shift α_g on the logit scale:

$$\sum_{i \in g} \tilde{w}_{di}\,\text{logit}^{-1}\left(\text{logit}(\hat{\theta}_{di}) + \alpha_g\right) = \tilde{T}_g, \tag{12}$$

  via robust one-dimensional root-finding, guaranteeing $\hat{\theta}^{\text{BM}}_{di} \in (0,1)$.

2.4 Official statistics reliability classification

NSOs classify domain estimates into publication tiers based on the Relative Standard Error ($\text{RSE} = \frac{\sqrt{\text{MSE}(\hat{\theta}_i)}}{|\hat{\theta}_i|} \times 100\%$). **fastsae** provides `add_reliability_flags()` implementing institutional thresholding: %

$$\text{Tier}_i = \begin{cases} \text{Reliable (Direct Publish)}, & \text{RSE}_i \le c_1, \\ \text{Use with Caution (Accompanying Warning)}, & c_1 < \text{RSE}_i \le c_2, \\ \text{Unreliable (Suppress or Aggregate)}, & \text{RSE}_i > c_2, \end{cases} \tag{13}$$

% where the default thresholds $(c_1, c_2) = (25\%, 50\%)$ follow the criteria of BPS-Statistics Indonesia (with $\text{RSE} \le c_1$ reliable and $\text{RSE} > c_2$ unreliable); other agencies' cut-offs can be supplied.

3 Software design and implementation

The architecture of **fastsae** unites compiled linear algebra with high-level Bayesian GMRF modeling and production reporting. Figure 1 outlines the structural flow.

The frequentist engines (`eblup_fh()`, `eblup_sfh()`, `eblup_stfh()`, `eblup_bhf()`, `eblup_twofold()`) never form a dense inverse of the marginal covariance: the spatial model works with sparse Cholesky factors (**RcppEigen**), the spatio-temporal and two-fold models with block-structured Woodbury identities, and the unit-level model with domain-level sufficient statistics, profiling $\boldsymbol{\beta}$ and σ^2_e out of the (restricted) likelihood. The bootstrap MSE loops run in parallel with OpenMP (`#pragma omp parallel for`); random numbers are drawn sequentially from R's generator beforehand, so results depend only on the seed and not on the number of threads. The Bayesian functions (`hb_area()`, `hb_unit()`,

fastsae High-Level Software Architecture

1. User-Facing Production API (R)		
Frequentist Modeling	**Bayesian HB Engine**	**Production Workflow**
`eblup_fh()`	`hb_area(...,family = "beta")`	`benchmark_sae()` (Two-Stage)
`eblup_sfh()`	`hb_area(...,family = "gamma")`	`compare_sae()`
`eblup_stfh()`	`hb_area(...,family = "poisson")`	`add_reliability_flags()`
`eblup_bhf()`	`hb_area(...,spatial = "bym2")`	`map_sae()` / `export_sae()`
2. Core Computational Backends		
Compiled C++17 Core	**INLA Deterministic Engine**	**Spatial Linear Algebra**
RcppArmadillo Matrix Algebra	Sparse Cholesky Factorization	Row-normalized Proximity
OpenMP Threaded Bootstrap	Full Laplace Approximation	PC Priors & GMRF Graphs
In-Place Fisher-Scoring Updates	Survey Dispersion Injections	Spectral Eigenvalue Bounds

Figure 1: Structural architecture and functional components of fastsae.

D	Execution Time: Median (seconds)			Memory Allocated (MB)		
	fastsae	sae	emdi	fastsae	sae	emdi
Fay-Herriot EBLUP (Classical)						
30	0.00055	0.00155	0.01088	0.339	0.153	4.18
100	0.00064	0.00344	0.05051	0.018	0.918	37.34
500	0.00107	0.19673	5.86366	0.085	18.239	880.31
1000	0.00119	1.50363	51.0224	0.503	71.977	3,773.60
Spatial Fay-Herriot (SEBLUP)						
30	0.00588	0.00410	0.01071	0.009	1.373	4.18
100	0.01388	0.07054	0.05299	0.021	17.976	37.34
500	0.21310	7.11793	3.13662	0.101	776.407	949.86
1000	0.99977	67.1518	51.0593	0.203	1,920.01	3,772.57
Spatio-Temporal Fay-Herriot (STEBLUP, $T = 5$)						
30	0.14272	0.11856	–	0.138	43.801	–
100	0.21415	17.9425	–	0.316	2,551.10	–
500	10.0430	253.343	–	1.633	8,124.37	–
1000	75.3340	1,786.49	–	3.293	32,478.3	–

Table 2: Frequentist runtime and memory allocation (point estimates and analytical MSE) across domain sample sizes (D). At $D = 1,000$, fastsae is $1,263\times$ faster than sae for FH, $67\times$ for SFH, and $24\times$ for ST-FH. At $D = 30$ the spatial and spatio-temporal fits are dominated by fixed costs, and sae is slightly faster.

`hb_twofold()`) translate formulas, spatial adjacency matrices $\boldsymbol{W}$, and survey dispersion arguments into **R-INLA** calls. How a model is estimated is specified separately from the model by an engine object: `inla_engine()` holds the INLA strategy, priors, and further `INLA::inla()` arguments, and `jax_engine()` fits `hb_area()` models by NUTS sampling in JAX/NumPyro (CPU or GPU) through reticulate. Standard S3 methods (`print()`, `summary()`, `plot()`, `coef()`, `fitted()`, `residuals()`) are implemented across all model classes.

4 Computational benchmarks and simulations

4.1 Frequentist scalability and memory efficiency: fastsae vs. sae and emdi

We evaluated computational speed and memory allocation across domain sizes $D = 30, 50, 100, 250, 500,$ and 1000 using simulated datasets with 5 auxiliary covariates. Times are medians measured with the bench package; memory is the total memory allocated by R during one call (`mem_alloc`), not the peak resident memory. The fastsae rows were measured with fastsae 1.0.0 under R 4.6.1 on an Apple M4 processor (10 cores); the sae and emdi rows come from an earlier run of the same design on different random draws. The scripts are in the `benchmarks/` directory of the package repository.

Table 2 demonstrates the scalability gains of fastsae. At $D = 1,000$, classical FH completes in about

D	Cross-Sectional Beta		Spatial Beta (Besag)		Spatio-Temporal Beta	
	fastsae	tipsae	fastsae	tipsae	fastsae	tipsae
30	1.20 s	2.35 s	1.22 s	25.72 s	1.48 s	540.57 s
50	1.23 s	3.77 s	1.23 s	65.72 s	1.71 s	908.17 s
100	1.26 s	8.24 s	1.25 s	100.02 s	–	–
250	1.32 s	26.28 s	1.33 s	155.90 s	–	–
500	1.40 s	64.21 s	1.46 s	248.82 s	–	–
1000	1.61 s	187.16 s	1.87 s	404.34 s	–	–

Table 3: Median execution time for Bayesian Beta models (spatio-temporal: $T = 5$). Both packages are fitted to the same simulated data. Point estimates of fastsae and tipsae for the cross-sectional model at $D = 1,000$ achieve $r = 0.9993$ concordance.

one millisecond in fastsae compared to 1.5 seconds in sae (1,263× slower) and 51 seconds in emdi (42,874× slower). For spatio-temporal models at $D = 1,000$ ($T = 5$), sae required nearly 30 minutes (1,786 seconds) and allocated 32.5 gigabytes, whereas fastsae finished in 75 seconds allocating 3.3 megabytes. **Operational Takeaway**: fastsae eliminates out-of-memory bottlenecks, enabling routine spatio-temporal estimation across thousands of domains on standard laptop hardware.

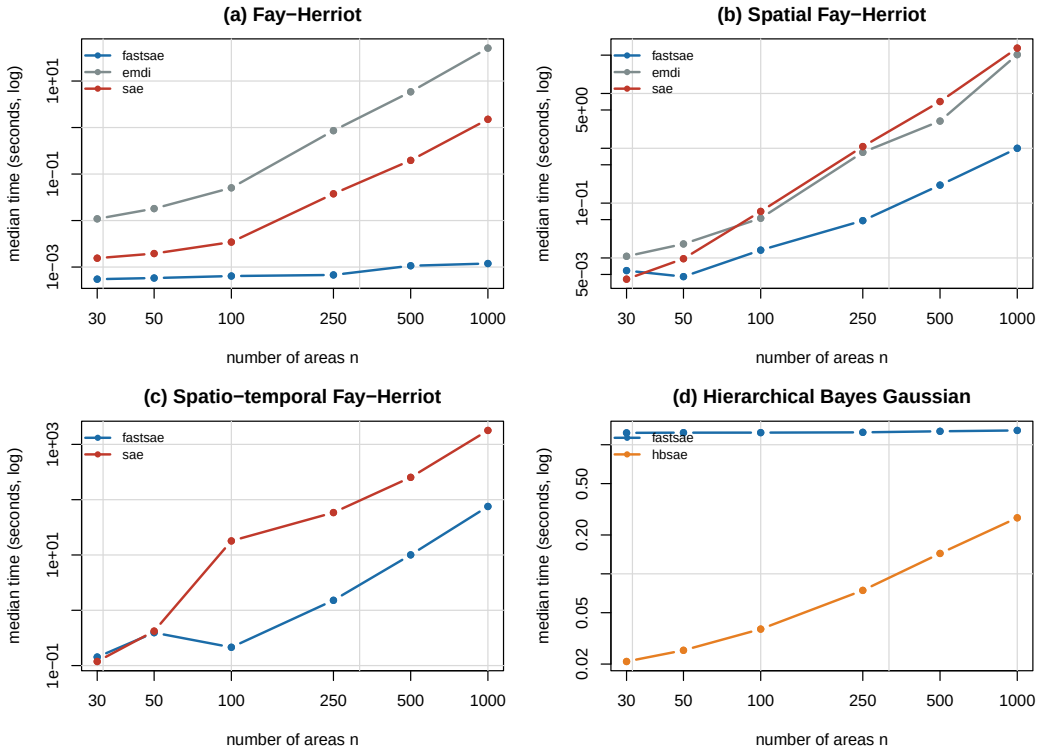


Figure 2: Median runtime scaling across domain sample sizes (D) on log-log scale for classical, spatial, spatio-temporal Fay-Herriot, and Bayesian Gaussian models.

4.2 Bayesian Beta Hierarchical Modeling scalability: fastsae vs. tipsae

For bounded rates, we benchmarked `hb_area(...,family = "beta")` against tipsae (De Nicolò and Gardini, 2022) using Stan (4 chains, 2,000 iterations).

As shown in Table 3, deterministic Laplace approximations in fastsae scale flatly with domain count: fitting a spatial Beta model at $D = 1,000$ requires only 1.87 seconds in fastsae compared to 404 seconds in tipsae (217× speedup). Most of the fastsae time is the fixed start-up cost of the INLA program (about 1.2 seconds even at $D = 30$). In spatio-temporal panel models ($D = 50, T = 5$), tipsae required 15.1 minutes (908 seconds), whereas fastsae completed in 1.71 seconds (530× speedup). Despite this speed differential, inferential concordance on $D = 1,000$ yields Pearson correlation $r = 0.9993$, Spearman $\rho = 0.9992$, and Mean Absolute Difference MAD $= 0.0197$. **Operational**

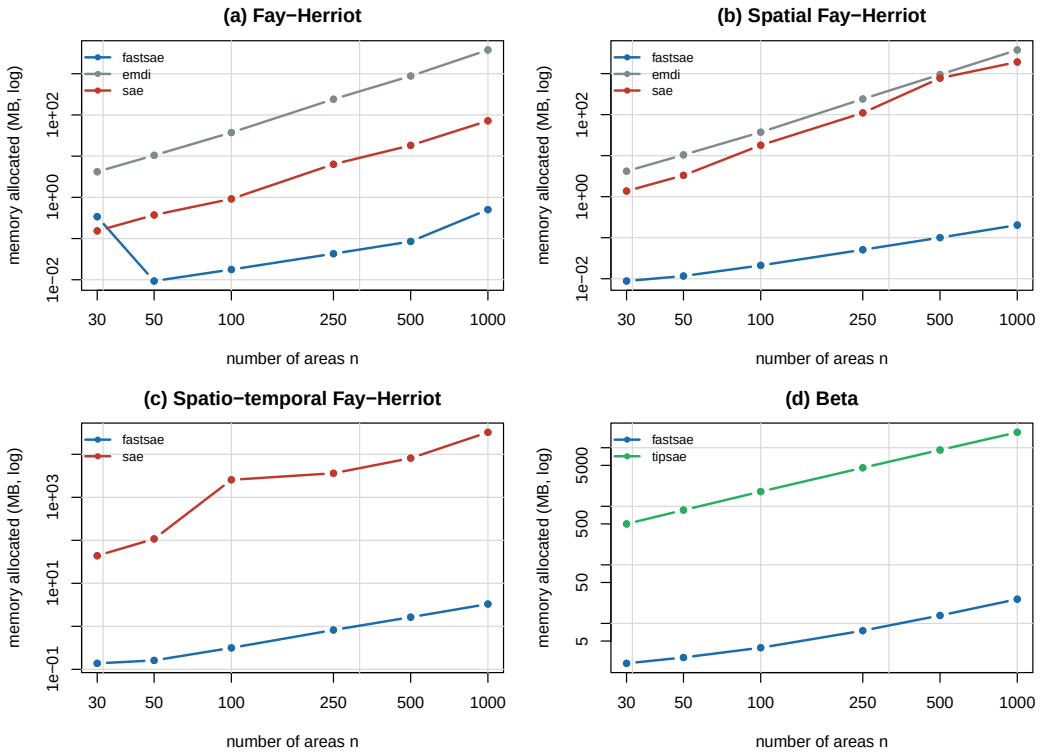


Figure 3: Memory allocated per call (MB, `bench::mark()$mem_alloc`) across domain counts (D) on log-log scale. **fastsae** avoids quadratic $O(D^2T^2)$ allocation.

Model	D	INLA	JAX (CPU)	JAX (Apple GPU)	**tipsae**
Gaussian	100	1.4 s	6.4 s	129 s	–
	500	1.4 s	45 s	155 s	–
	1,000	1.5 s	94 s	205 s	–
	2,500	2.5 s	713 s	474 s	–
Beta	100	1.2 s	6.0 s	110 s	107 s
	500	1.4 s	36 s	136 s	268 s
	1,000	1.8 s	78 s	201 s	401 s
	2,500	3.2 s	538 s	393 s	946 s

Table 4: Warm runtime of `hb_area()` with an ICAR area effect by estimation engine on an Apple M4. JAX: NumPyro NUTS, 2 chains × 1,000 draws after 500 warmup. **tipsae**: Stan defaults, Beta model only.

Takeaway: Deterministic Laplace approximation in **fastsae** achieves inferential equivalence with fully converged Hamiltonian MCMC in seconds rather than minutes or hours.

4.3 Estimation engines: INLA versus MCMC on CPU and GPU

The same `hb_area()` model can be estimated by INLA (`inla_engine()`) or by full MCMC with the No-U-Turn sampler of NumPyro on JAX (`jax_engine()`), on the CPU or on a GPU. We compared the engines on spatial models with a Besag (ICAR) area effect on a regular grid with queen adjacency, for a Gaussian Fay-Herriot model and a Beta model; for the Beta model, **tipsae** was fitted with its ICAR spatial error. Each engine ran in its own R process on an Apple M4 (10 cores, 16 GB), JAX with 2 chains of 1,000 draws after 500 warmup, and each fit was run twice on the same data; Table~4 reports the second (warm) run.

INLA is the fastest engine at every size: at $D = 2,500$ it fits the spatial Beta model in 3.2 s, 296 times faster than **tipsae**. Among the MCMC options, JAX on the CPU is 2 to 18 times faster than Stan, with the gap narrowing as D grows, because the JAX engine represents the ICAR effect through the eigenvectors of the $D \times D$ graph Laplacian ($O(D^2)$ work per gradient). The Apple GPU carries a fixed cost of about two minutes per fit, since each step of the NUTS tree-building loop synchronises the GPU

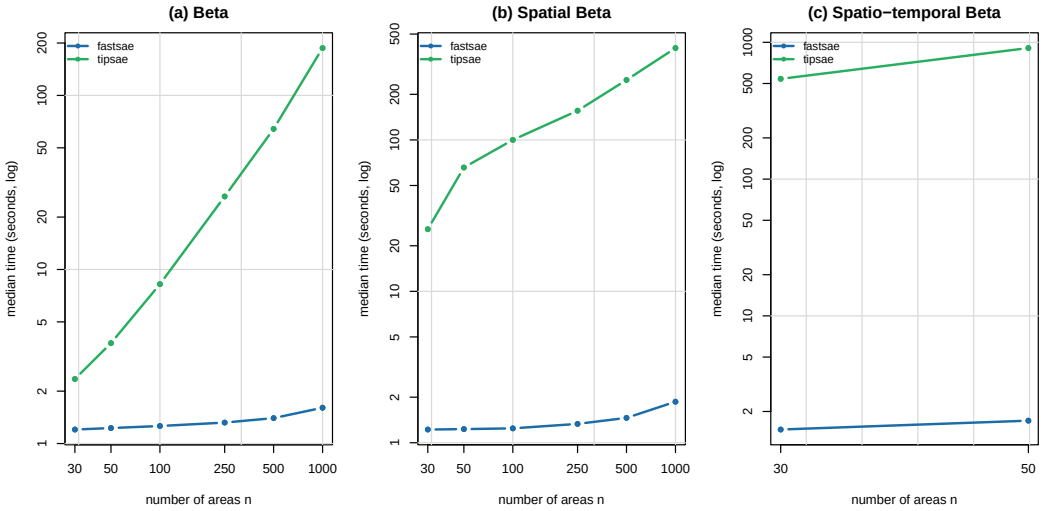


Figure 4: Runtime scaling across domain sample sizes (D) for cross-sectional, spatial, and spatio-temporal Bayesian Beta models.

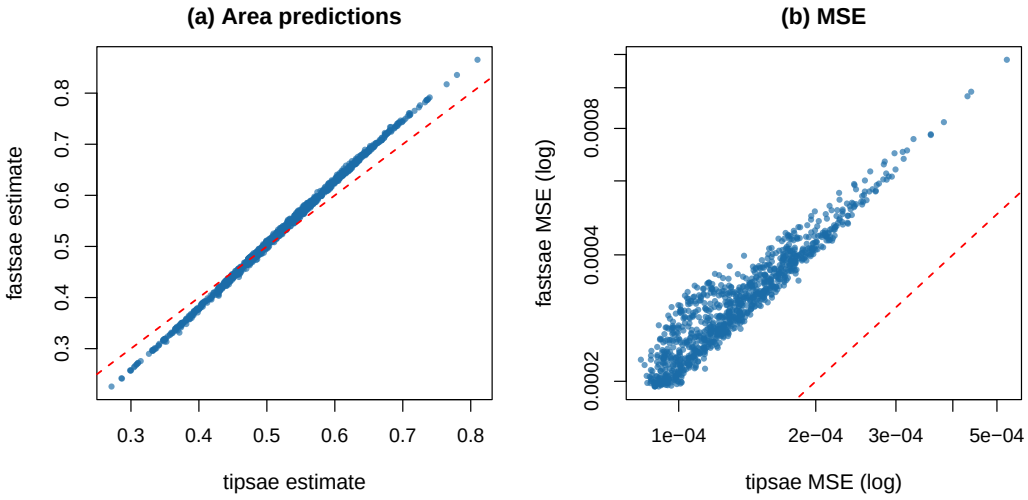


Figure 5: Numerical equivalence and inferential concordance between **fastsae** (INLA) and **tipsae** (Stan HMC) at $D = 1{,}000$ ($r = 0.9993$).

Estimator	Emp. RMSE	RRMSE (%)	MARB (%)	95% CrI Cov.	CrI Width	Runtime
Direct Sample Estimator	0.0719	27.1%	21.7%	–	–	–
fastsae (Spatial Besag)	**0.0481**	**19.1%**	16.9%	89.0%	**0.173**	**4.17 s**
tipsae (Spatial Besag)	0.0487	18.7%	**15.7%**	**93.7%**	0.202	36.86 s

Table 5: Finite population Monte Carlo simulation results ($D = 500$, $R = 25$). fastsae achieves an efficiency gain of 1.92× over the direct survey estimator.

and the CPU, but it overtakes the CPU at $D = 2,500$ for both models. Running NumPyro's NUTS on Apple GPUs required two workarounds in fastsae, because **jax-metal** cannot compile the `popcnt` bit count or `lax.cond` statements with constant outputs; both are replaced by equivalent operations when `device = "metal"`. The engines agree closely: the posterior means of JAX on either device correlate with INLA at $r \geq 0.9999$, and the largest split $\hat{R}$ over all JAX fits is 1.007 (1.083 for tipsae with its default run length).

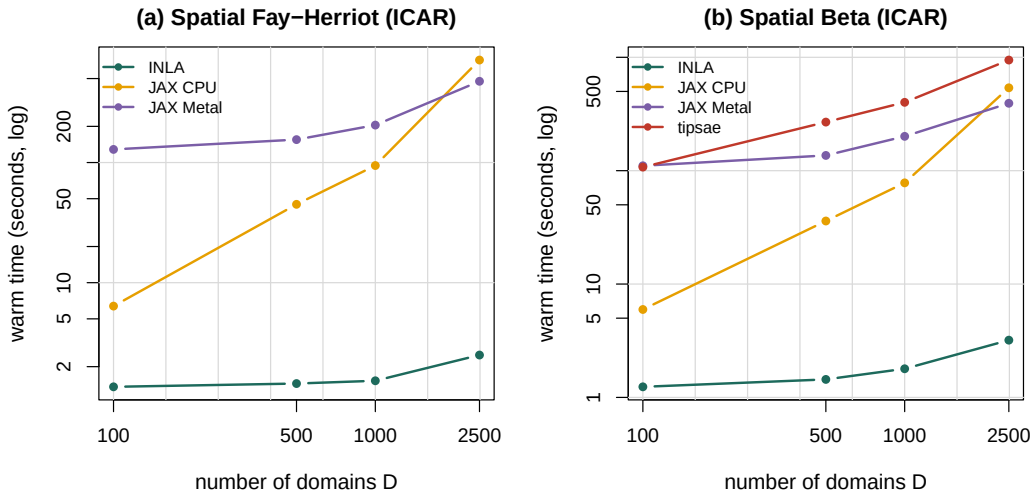


Figure 6: Warm runtime of the `hb_area()` estimation engines for spatial (ICAR) Gaussian and Beta models on an Apple M4 (log scales).

4.4 Monte Carlo finite population validation against ground truth

To assess frequentist design properties, we conducted design-based Monte Carlo simulations under known ground truth at national scale ($D = 500$ small areas, $R = 25$ independent sample draws) using a synthetic finite population of $N = 250,000$ individuals.

Table 5 demonstrates that `hb_area()` reduces empirical RMSE from 0.0719 (direct survey estimator) to 0.0481, yielding a 1.92× empirical efficiency gain. Relative Root Mean Squared Error drops from 27.1% to 19.1%. Comparing Bayesian engines, fastsae achieves a slightly lower empirical RMSE than tipsae (0.0481 vs. 0.0487) and produces narrower credible intervals (0.173 vs. 0.202) while requiring nearly 9× less runtime per fit. **Operational Takeaway**: fastsae produces well-calibrated, highly efficient area-level predictions that closely track true finite-population parameters while scaling effortlessly to national-scale administrative systems.

5 Applied illustration and production workflow

We demonstrate an end-to-end NSO production workflow using the benchmark dataset `sim_area` ($D = 42$ domains across 6 provincial groups).

5.1 Loading data, spatial weights, and fitting models

```
library("fastsae")
data("sim_area")
sim_area$prov <- rep(paste0("Prov_", 1:6), each = 7)
```

```
sim_area$pop <- round(runif(nrow(sim_area), 15000, 45000))

# Construct row-standardized spatial proximity matrix
coords <- cbind(sim_area$x_coord, sim_area$y_coord)
dists <- as.matrix(dist(coords))
W <- (dists < quantile(dists, 0.18)) * 1
diag(W) <- 0
rs <- rowSums(W); rs[rs == 0] <- 1; W_std <- W / rs

# Fit Frequentist and Bayesian Models
fit_fh <- eblup_fh(y_gaussian ~ x1 + x2, vardir = "vardir",
  data = sim_area, method = "REML", print_result = FALSE)
fit_sfh <- eblup_sfh(y_gaussian ~ x1 + x2, vardir = "vardir",
  domain = "area", W = W_std, data = sim_area,
  method = "REML", print_result = FALSE)
fit_beta <- hb_area(y_beta ~ x1 + x2, vardir = "vardir",
  data = sim_area, family = "beta", spatial = "bym2",
  W = W, print_result = FALSE)
```

5.2 Automated model concordance evaluation

Practitioners can immediately compare competitive models via `compare_sae()`:

```
comp <- compare_sae(fit_fh, fit_sfh, names = c("FH", "SFH"))
comp$metrics[c(1, 2, 4, 6, 8, 9), ]
```

```
#>                                                Metric  Value
#> 1                                   Matched Domains     42
#> 2                            Pearson Correlation (r) 0.9996
#> 4                      Mean Absolute Difference (MAE) 0.0279
#> 6                           Mean MSE Ratio (FH / SFH) 0.9897
#> 8                      Domains where SFH has lower MSE 3 / 42
#> 9 Percentage of domains where SFH is more efficient (%)   7.1%
```

5.3 Two-stage hierarchical benchmarking calibration

To harmonize domain estimates with an official national poverty target ($T_{\text{nat}} = 0.50$), we apply two-stage calibration:

```
bm <- benchmark_sae(fit_fh, national_target = 0.50, group = "prov",
  weight = "pop", method = "ratio", type = "mean")
head(bm[, c("domain", "group", "original", "benchmarked", "rel_adjustment_pct")], 4)
```

```
#> === fastsae Small Area Benchmark Calibration ===
#> Method: Ratio (Multiplicative Adjustment)
#> Target Type: Weighted Mean / Rate
#> Total Domains: 4
#>
#> First 6 benchmarked domains (Level 2):
#>   domain  group   original benchmarked rel_adjustment_pct
#> 1      1 Prov_1  2.6646558   0.9111240          -65.80707
#> 2      2 Prov_1 -1.6547233  -0.5657984          -65.80707
#> 3      3 Prov_1  1.2504674   0.4275715          -65.80707
#> 4      4 Prov_1 -0.7988192  -0.2731397          -65.80707
```

The calibrated estimates satisfy $\sum(w_i / \sum w_i)\hat{\theta}_i^{\text{BM}} = 0.50$ exactly.

5.4 Official statistics reliability classification

Using `add_reliability_flags()`, domains are classified into official dissemination tiers:

```
flags <- add_reliability_flags(fit_beta)
table(flags$reliability)
```

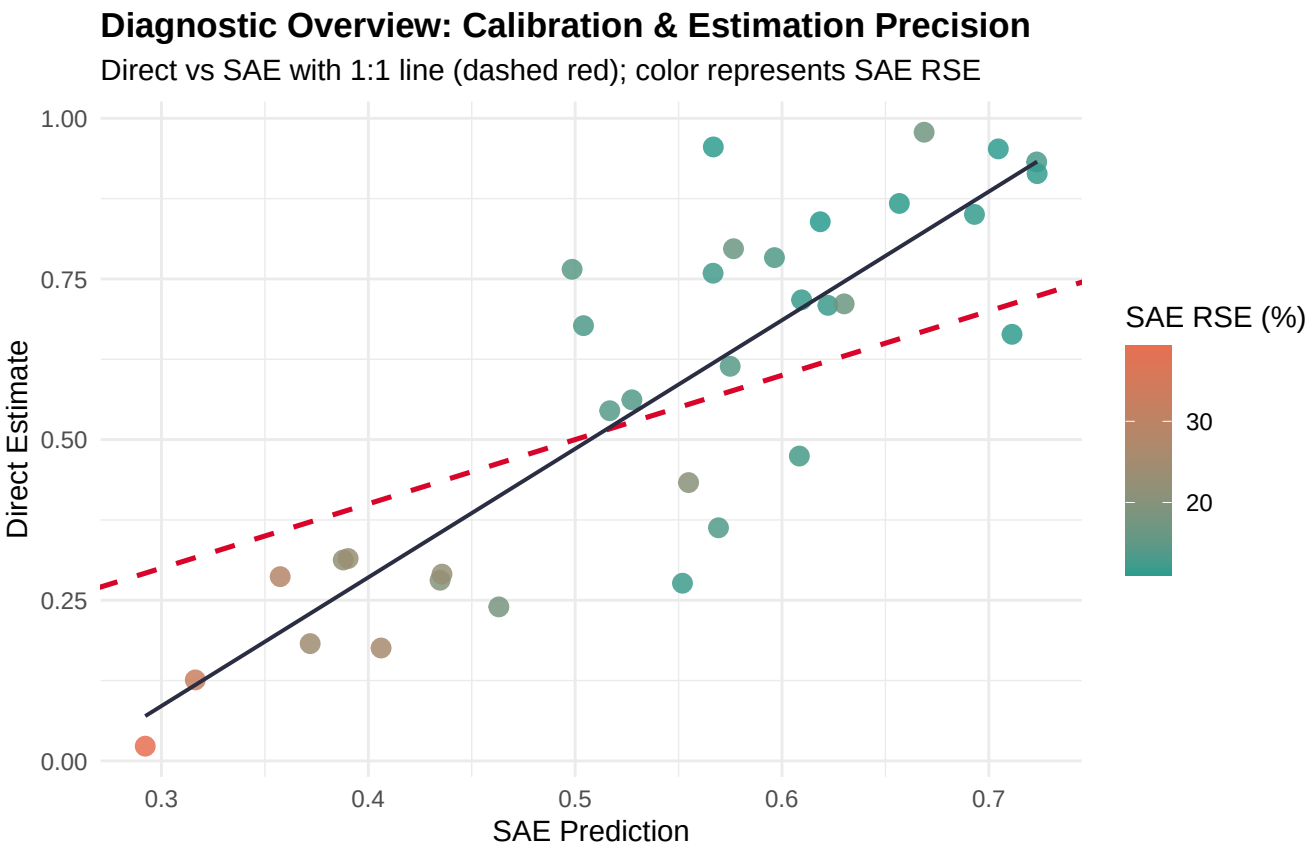


Figure 7: Model diagnostic panels generated via autoplot.

```
#>
#>               Reliable (RSE <= 25%) Use with caution (25% < RSE <= 50%)
#>                                  37                                     5
#>             Unreliable (RSE > 50%)
#>                                   0
```

5.5 Model diagnostics and visualization

Diagnostics via `diagnose()` and `autoplot()` provide visual quality verification:

```
diag_beta <- diagnose(fit_beta)
autoplot(diag_beta)
```

6 Summary and discussion

Small area estimation in official statistics requires reconciling mathematical rigor with strict computational and operational constraints. **fastsae** resolves these challenges by bridging high-performance compiled linear algebra, deterministic Bayesian approximations, and production-ready dissemination tools.

As synthesized in Table 6, **fastsae** provides substantial efficiency gains across both frequentist and Bayesian paradigms. Future developments will focus on expanding unit-level spatial EBP under non-Gaussian likelihoods, integrating zero-inflated Beta (ZIB) specifications for extreme sparse indicators, and supporting direct spatial polygon streaming.

Computational details

The **fastsae** results in this paper were obtained using R 4.6.1 with **fastsae** 1.0.0, **R-INLA** 26.8.7, and **ggplot2** 4.0.3 on an Apple M4 processor; the comparison runs used **sae** 1.3, **emdi** 2.2, **tipsae** 1.0,

Dimension	Observed Empirical Evidence	Institutional Implication ("So What")
Frequentist Scalability	$1,263\times$ speedup for FH and $67\times$ for SFH at $D = 1,000$; a few MB allocated for ST-FH (vs 32 GB in **sae**).	Eliminates server memory exhaustion, allowing desktop computers to execute nationwide bootstrap MSE routines.
Bayesian EBP via INLA	$117\times$ to $530\times$ speedup over Stan MCMC at the largest benchmarked sizes; point correlation $r = 0.9993$ on $D = 1,000$.	Eliminates manual MCMC chain diagnostics and divergent transition failures in routine production timelines.
Design-Based Ground Truth	$1.92\times$ empirical efficiency gain over direct estimator; 89% empirical coverage.	Delivers substantial MSE reductions while maintaining nominal interval coverage under repeated sampling.
Hierarchical Benchmarking	Simultaneous Level 0 and Level 1 target alignment with logit boundary safety.	Guarantees numeric consistency with published aggregates, eliminating public confusion without violating probability bounds.
Turnkey Dissemination	Direct RSE classification, spatial choropleths, and multi-sheet exports.	Collapses multi-script analyst workflows into standardized, reproducible publication pipelines.

Table 6: Synthesis of empirical findings and operational implications of **fastsae**.

and **hbsae** 1.2. R and packages from CRAN are available at https://CRAN.R-project.org/. **R-INLA** is available from https://www.r-inla.org/. The **fastsae** package is available on GitHub at https://github.com/ridsonap/fastsae.

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